Thalassemia

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Introduction

- Since globin is a protein, it is encoded by genes located on certain chromosomes within the genetic material and these genes are transcribed and translated to make proteins.

- **Thalassemias:** a group of genetic or inherited diseases caused by genetic mutations and characterized by decreased or absent synthesis of globin chains of hemoglobin (quantitative disorder).

- **Hemoglobinopathies:** are a group of genetic or inherited diseases characterized by defects in the structure of globin chains of hemoglobin (qualitative or structural abnormalities).

- Thalassemias are the most common single-gene disorder in humans.

- Thalassemia results from a reduced or absent synthesis of one or more globin chains.

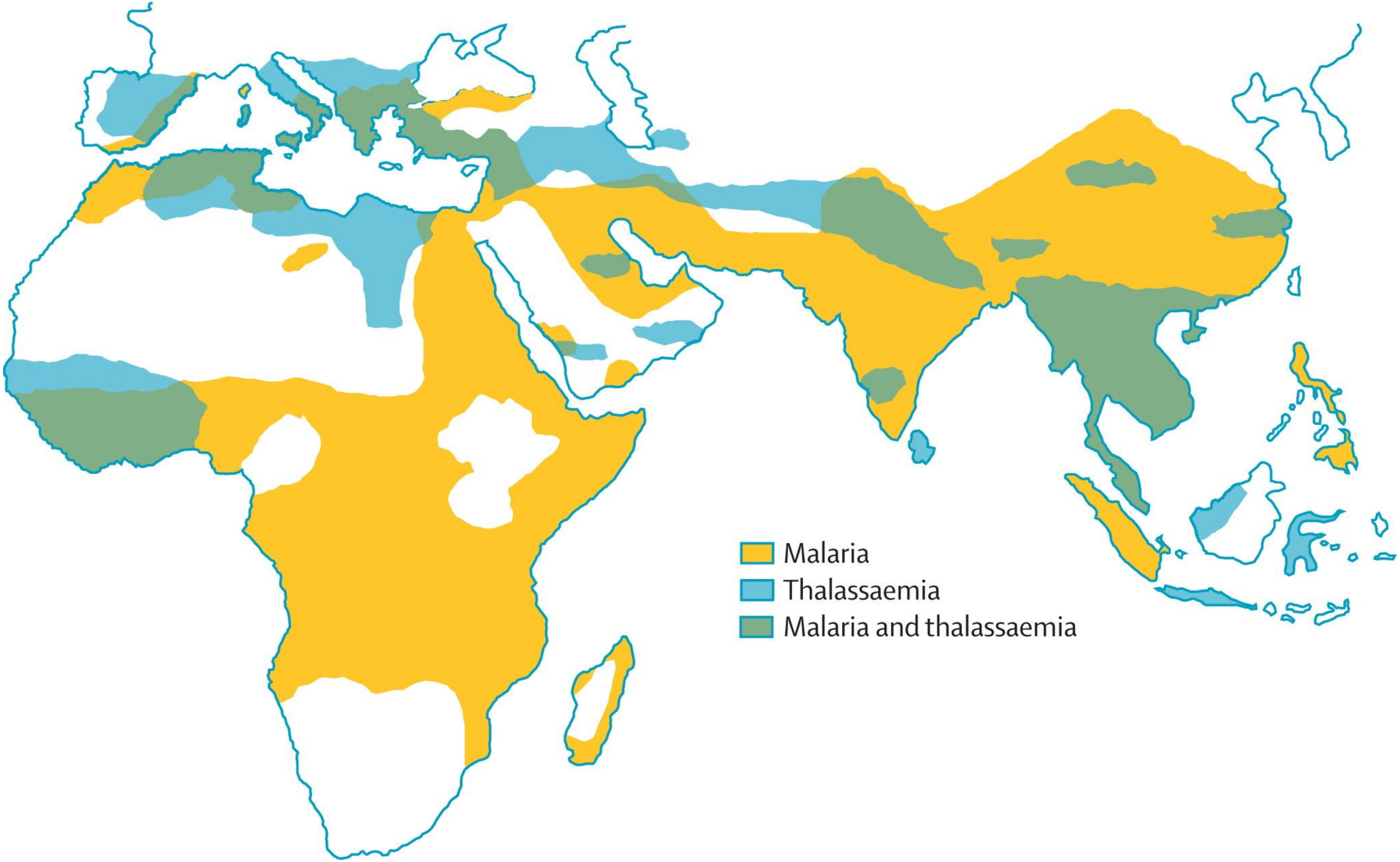
- This decreased or absent globin production leads to decreased Hb production, imbalanced α /β chain synthesis, and damage to RBCs or their precursors by the excess accumulation of the globin chain that isn’t affected.

- Thalassemias are named according to the chain with reduced or absent synthesis **(α-thalassemia and β-thalassemia).**

- Approximately, 7% of the world’s population has a thalassemia or hemoglobinopathy.

Epidemiology

- Annually an estimated 56,000 infants are born with a form of thalassemia major.
- The distribution is concentrated in the “**thalassemia belt**” that extends from the Mediterranean east through the Middle East and India to Southeast Asia and south to northern Africa.
- Sardinia, Cyprus, and Greece have the highest frequency in Europe (6-19%).
- The thalassemia belt occurs simultaneously with areas in which malaria is prevalent.
- Thalassemia minor appears to impart resistance to malaria.


Malaria
Thalassaemia
Malaria and thalassaemia

Categories of Thalassemia

- Thalassemia can be categorized according to the defective chain.

- Thalassemia is caused by non-deletional or missing (deletional) mutations.

- If the defective chain is β, so, you will have β-thalassemia.

- If the defective chain is α, so, you will have α-thalassemia.

- The genotype for normal β chain synthesis is designated β/β.

- In the β-thalassemia, β^o is the designation for mutations in the β gene in which no β chains are produced.

- β^o/β^o (homozygous) doesn't produce HbA ($\alpha_2\beta_2$).

- β+ is the designation for various mutations in the β gene that result in a partial deficiency of β chains (5-30% of normal) which results in a decreased production of HbA.

- There are 4α genes responsible for the synthesis of α chain (a normal genotype is designated $\alpha\alpha/\alpha\alpha$).

- α+ is used to indicate a deletion of either the α1 or α2 gene in the α gene complex, which results in a decreased production of α chains from that chromosome.

- α^{o} is used to indicate deletion of both the α1 and α2 gene in the α gene complex, which results in no production of α chains from that chromosome.

Genetic defect causing Thalassemia

- Types of genetic defects that cause a decrease or lack of production of a particular globin chain and thalassemia include single nucleotide (point) mutations, small insertions or deletions, or large deletions.

- The mechanisms by which these mutations interfere with globin chain production include:

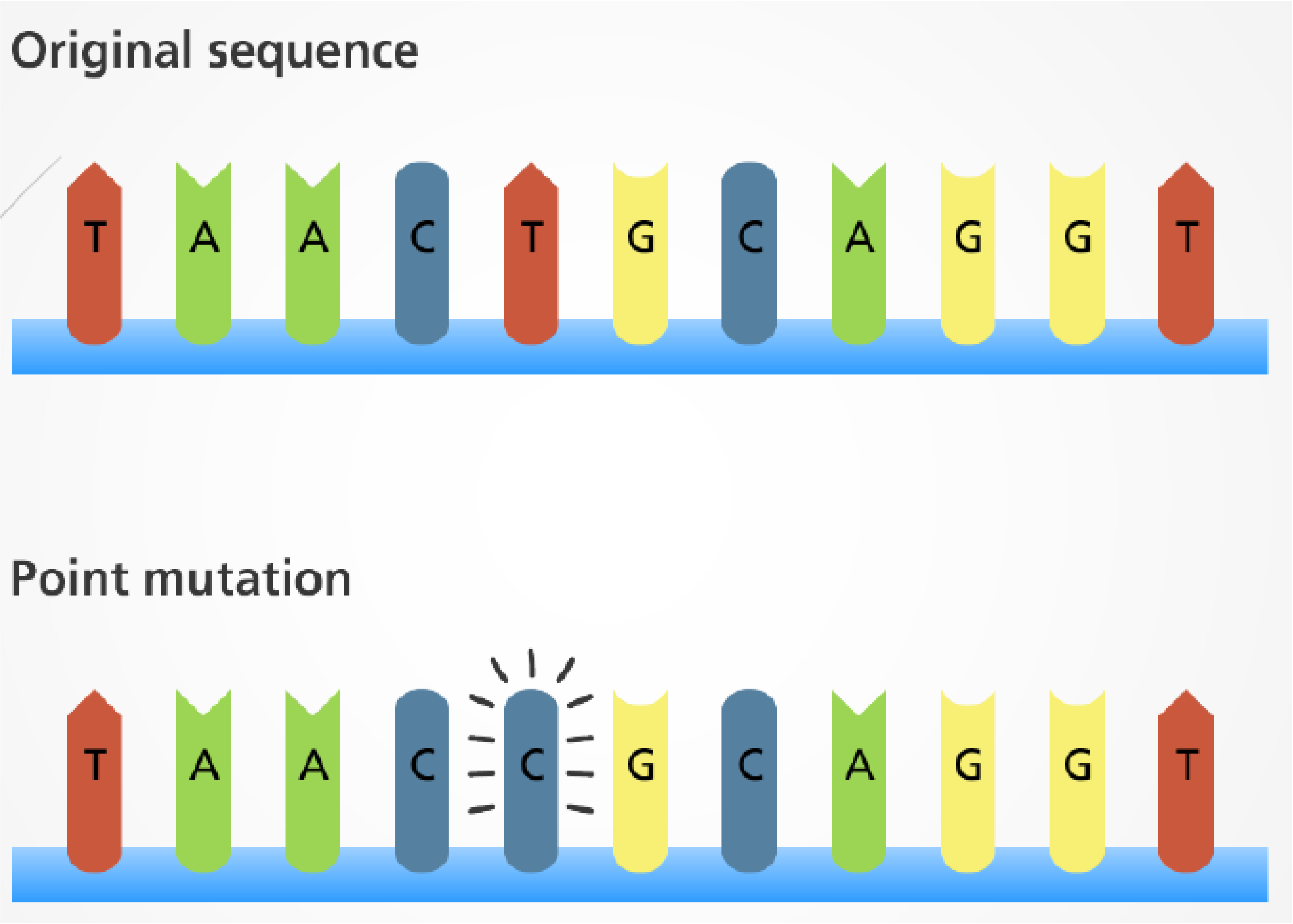
  - Reduced or absent transcription of mRNA.

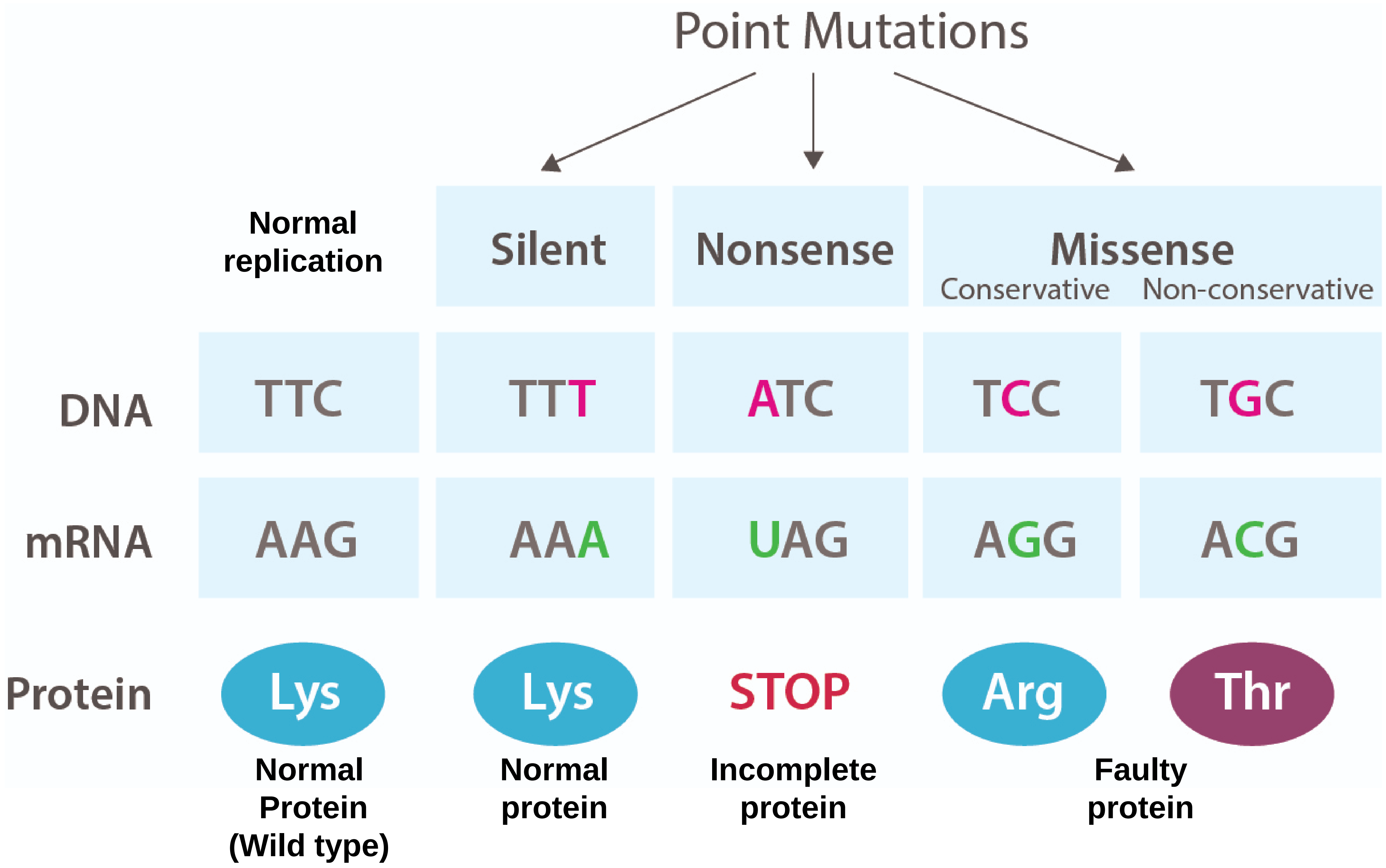
  - mRNA processing errors due to mutations that add or remove splice sites resulting in no or altered chain production.

- Translation errors due to mutations that change the codon reading frame (frameshift mutations), substitute an incorrect amino acid codon (missense mutations), add a stop codon causing premature chain termination (nonsense), or remove a stop codon, which results in an elongated and unstable mRNA that produces a dysfunctional globin chain.

- Deletion of one or more globin genes resulting in the lack of production of the corresponding globin chains.

➢ All of these heterogeneous genetic mutations cause a reduction or lack of synthesis of one or more globin chains, resulting in the thalassemia syndromes.


Original sequence
T A A C T G C A G G T
Point mutation
T A A C C G C A G G T


Point Mutations
Normal replication
Silent
Nonsense
Missense
Conservative
Non-conservative
DNA
TTC
TTT
ATC
TCC
TGC
mRNA
AAG
AAA
UAG
AGG
ACG
Protein
Lys
Lys
STOP
Arg
Thr
Normal Protein (Wild type)
Normal protein
Incomplete protein
Faulty protein

- **Thalassaemia:** is a genetic disorder resulting from a reduced rate of synthesis of α or β chains.

Clinically classifications

1. The main syndromes are transfusion-dependent and called thalassemia **major**

2. Non-transfusion dependent thalassemia **(thalassemia intermedia)** with a moderate degree of anemia due to a variety of genetic defects

3. **Thalassaemia minor**, usually due to a carrier state for α- or β-thalassaemia.

Or classifications according to the defective globin gene:

A. A-thalassemia
B. β-Thalassaemia

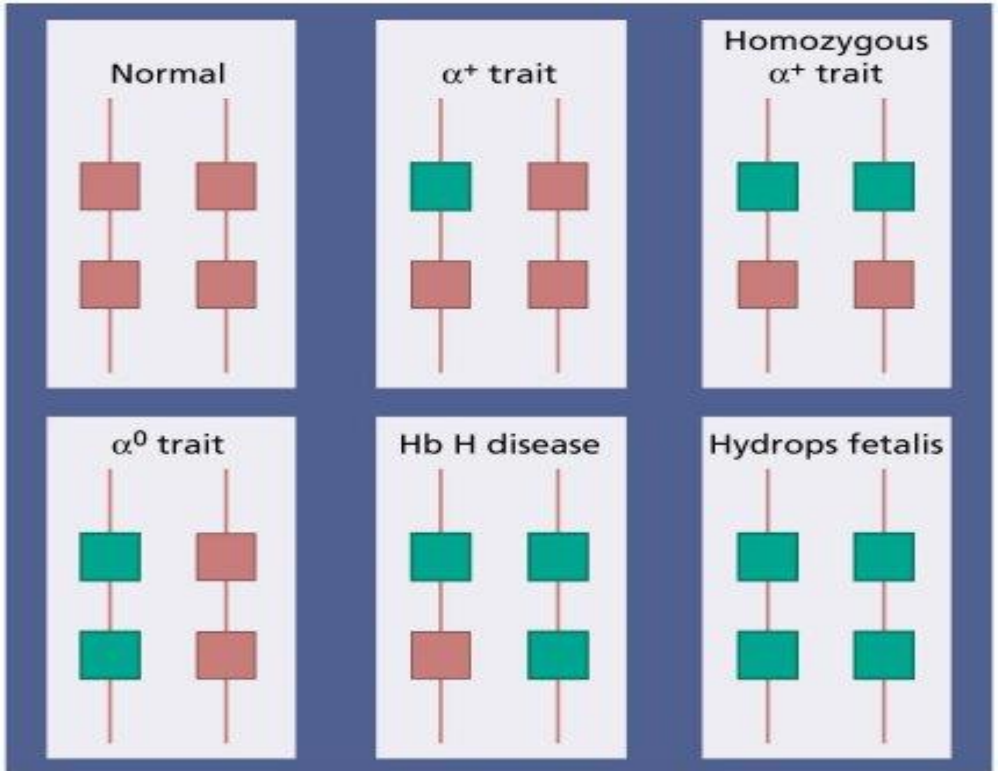
- β-Thalassaemia is more common in the Mediterranean region while α-thalassaemia is more common in the Far East.

A-thalassemia

- These are caused by α-globin gene deletions or less frequently mutations.

- The clinical severity is related to the number of the four α-globin genes missing or inactive.

- Loss of all four genes completely suppresses α-chain synthesis and because the α-chain is essential in fetal as well as in adult hemoglobin **this is incompatible with life and leads to death** *in utero* ***(the disease called hydrops fetalis).***



Hydrops fetalis: Hb parts (γ4)

- ***Three α gene deletion leads*** to a moderately severe (hemoglobin 70–110 g/L) microcytic, hypochromic anemia with splenomegaly. This is known as **Hb H disease because hemoglobin H (β4) can be** detected in the red cells of these patients by electrophoresis or in reticulocyte preparations. In fetal life, we will see Hb Barts (γ4).

- The α-thalassemia **traits are caused by the loss of one or two genes** and are usually not associated with anemia. The mean corpuscular volume (MCV) and mean corpuscular hemoglobin (MCH) are low and the red cell count is over 5.5×10^{12}/L. Haemoglobin electrophoresis is normal and DNA analysis is needed to be certain of the diagnosis.

β-Thalassaemia syndromes

B-thalassemia **major**

- This condition occurs on average in one of four offspring if both parents are carriers of the β-thalassemia trait.

- Either no β chain (β^{o}) or small amounts (β+) are synthesized.

- Excess α chains precipitate in erythroblasts and mature red cells causing severe ineffective erythropoiesis and hemolysis that are typical of this disease.

- The greater the α-chain excess, the more severe the anemia.

- Production of γ chains helps to ‘mop up’ excess α chains and to ameliorate the condition.

- Over 400 different genetic defects have been detected.

- Unlike α-thalassaemia, the majority of genetic lesions are point mutations rather than gene deletions.
- These mutations may be within the gene complex itself or in promoter or enhancer regions.
- Certain mutations are particularly frequent in some communities and this may simplify antenatal diagnosis aimed at detecting the mutations in fetal DNA.

Clinical features of thalassemia

1. **Severe anemia: becomes apparent at 3–6 months after** birth when the switch from γ- to β-chain production should take place. Pallor and a swollen abdomen.

2. **Enlargement of the liver and spleen occurs as a result of** excessive red cell destruction, and later because of iron overload.

The large spleen increases blood requirements by increasing red cell destruction and pooling, and by causing expansion of the plasma volume.

3. **Expansion of bones caused by intense marrow hyperplasia** leads to thalassaemic facies and thinning of the cortex of many bones with a tendency to fractures.

The facial appearance of a child with β-thalassemia major. The skull is bossed with prominent frontal and parietal bones; the maxilla is enlarged.

4. **Thalassaemia major is the disease that most frequently underlies transfusional iron overload.**

- **Regular transfusions** are usually commenced in the first year of life and unless the disease is cured by stem cell transplantation, are continued for life.

- Iron absorption is increased because of low serum hepcidin

- In children, failure of growth and delayed puberty are frequent, and without iron chelation, death from cardiac damage usually occurs in teenagers.

5. **Infections occur frequently.**

- **In infancy, without adequate** transfusion, anemia predisposes to bacterial infections. Pneumococcal, *Haemophilus, and meningococcal infections* are likely if splenectomy has been carried out.

- *Transfusion* of viruses by blood transfusion may occur. As a result of the reduction of deaths from cardiac iron overload by improved chelation therapy, infections now account for an increasing proportion of deaths in thalassemia major.

6. **Liver disease in thalassemia is most frequently**

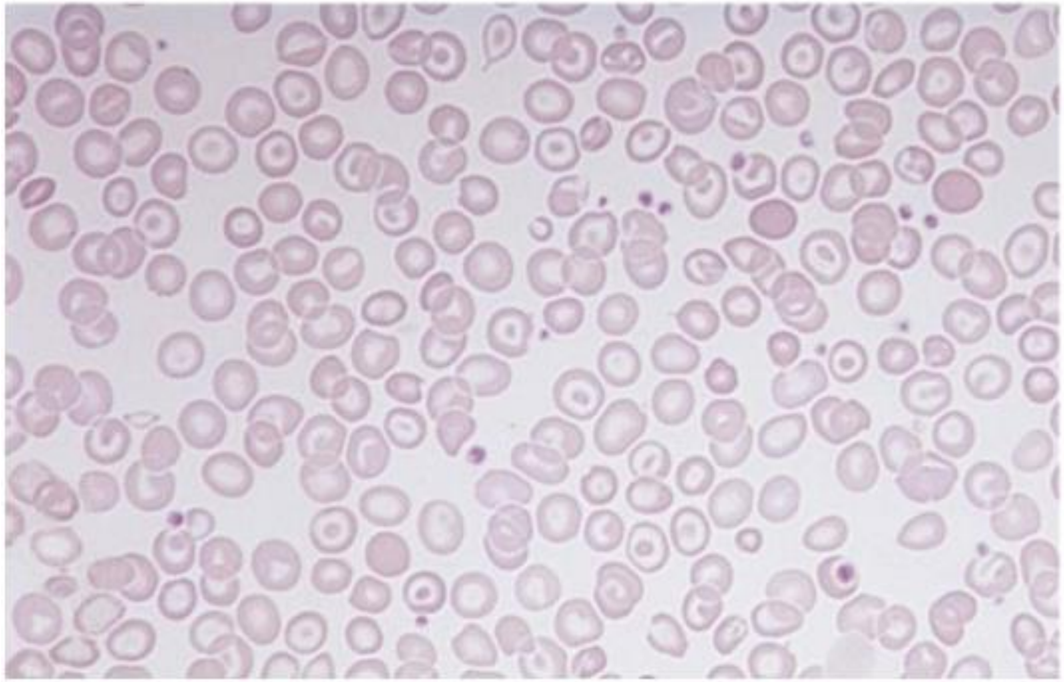
- **a result of** hepatitis C but hepatitis B is also common where the virus is endemic. Human immunodeficiency virus (HIV) has been transmitted to some patients by blood transfusion.

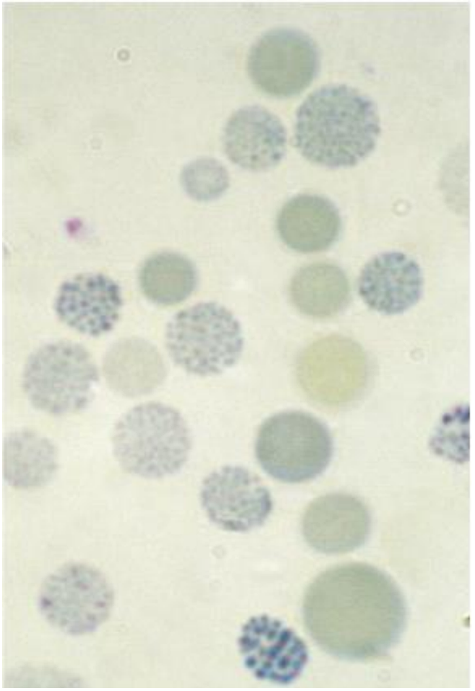
- Iron overload may also cause liver damage.

7. **Osteoporosis:** A systemic skeletal condition marked by reduced bone density and deterioration of bone tissue structure, resulting in increased porosity and a heightened risk of fractures. It is the leading cause of bone fractures in older adults.

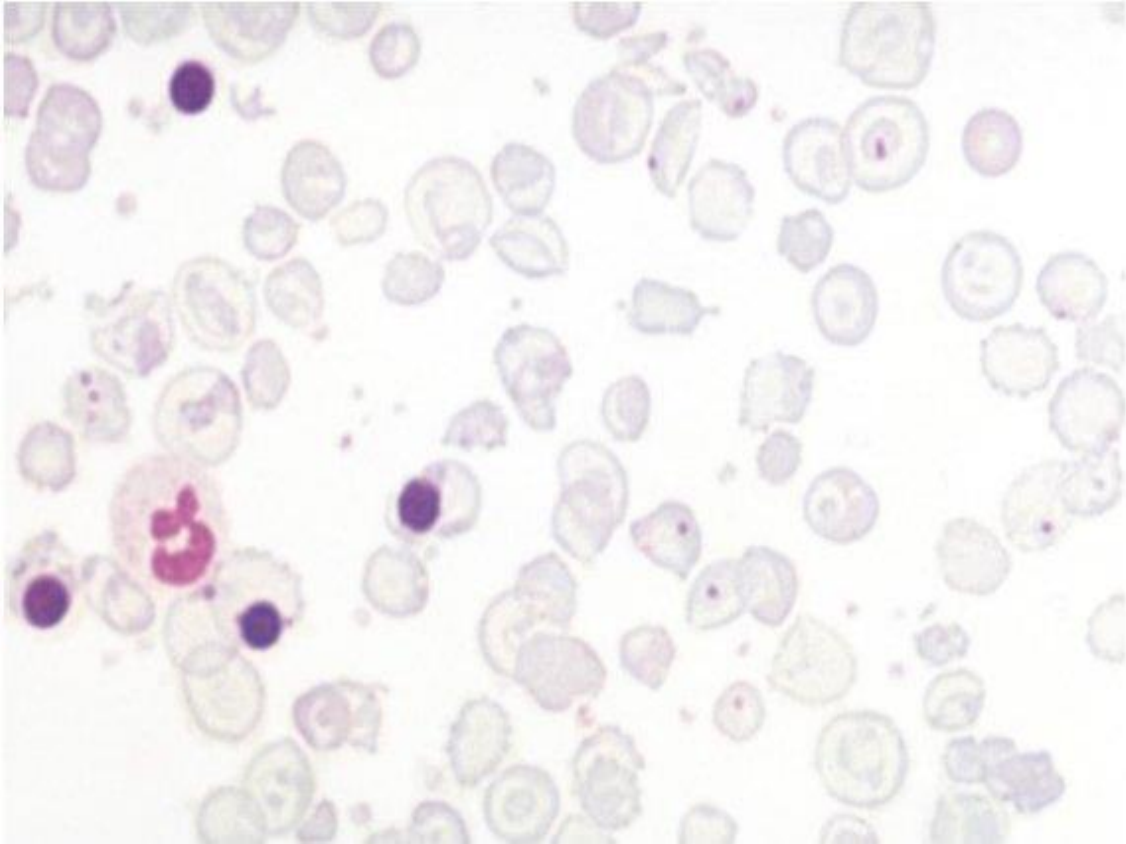
Laboratory diagnosis

1. **There is a severe hypochromic, microcytic anemia with** normoblasts, target cells, and basophilic stippling in the blood film.





Supravital staining with brilliant cresyl blue reveals multiple fine, deeply stained deposits (‘golf ball’ cells) caused by precipitation of aggregates of β-globin chains.



B-thalassemia blood film showed:

1. Hypochromic microcytic cells,
2. Target cells and many nucleated red cells (normoblast).
3. Holly-Jolly bodies (nuclear remnants (clusters of DNA) in circulating erythrocytes).

2. β*-Thalassaemia trait (minor)*

- This is a common, usually symptomless, abnormality characterized by α-thalassemia trait by a hypochromic, microcytic blood picture (MCV and MCH very low) but high red cell count ($>5.5 \times 10^{12}$/L) and mild anemia (hemoglobin 100–120 g/L).

- It is usually more severe than α thalassemia trait. A raised Hb A2 (>3.5%) confirms the diagnosis.

- The diagnosis allows the possibility of prenatal counseling. If the partner also has β-thalassemia trait there is a 25% risk of a thalassemia major child.

3. Non-transfusion dependent thalassemia (thalassemia intermedia)

- This is thalassemia of moderate severity (hemoglobin 70–100 g/L) without the need for regular transfusions.

- It is a clinical syndrome caused by a variety of genetic defects:
  A. Homozygous β-thalassemia with production of more Hb F than usual, e.g. due to mutations of the *BCL11A* gene.

  B. or with mild defects in β-chain synthesis, by β-thalassemia trait alone of unusual severity (‘dominant’ β-thalassemia).

  C. or by β-thalassemia trait in association with mild globin abnormalities such as Hb Lepore (δβ-fusion chain).

Treatment:

1. Blood transfusion: 2-3 units every 4-6 weeks
2. Folic acid 5mg/day
3. Iron chelating therapy
4. Endocrine therapy (in case of puberty and DM)
5. Immunization against hepatitis B
6. Stem cell transplantation 80-90% success rate in younger individuals.

Patient 1

Hemoglobin	Result	Normal
Hb F	80%	≤ 1%
Hb A	Absent	80-90%
Hb A2	9%	1-3

Beta thalassaemia major

Patient 2

Hemoglobin	Result	Normal
Hb F	9%	≤ 1%
Hb A	50%	80-90%
Hb A2	15%	1-3

Beta thalassaemia trait